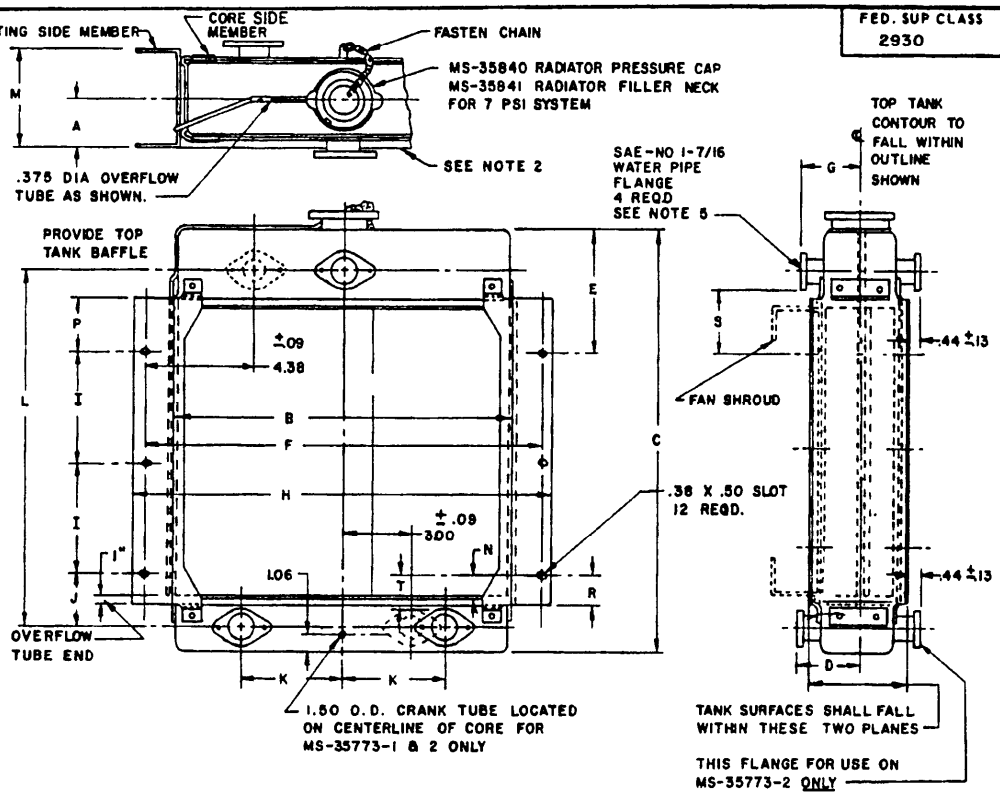
Review Activity: MO 82
User Activity: YD, MC

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MS PART NO	A ±.13	B MAX	C ±.13	D ±.13	E ±.13	F ±.13	G ±.13	H ±.13	I
MS-35773-1	2.22	15.32	20.13	2.18	4.63	16.76	2.81	18.00	6.00
MS-35773-2	1.81	18.32	23.00	2.81	4.75	19.25	2.81	20.50	7.38
MS-35773-3	1.81	22.32	30.00	2.81	5.50	23.38	2.81	24.63	10.50

J ±.09	K ±.13	L ±.13	M ±.06	N ±.13	P MAX	R MAX	FAN SHROUD S MAX	FAN SHROUD T MAX	U ±.13
2.25	5.00	16.81	4.44	0.88	1.38	1.38	1.47	1.16	13.94
2.38	5.13	20.13	4.44	0.88	2.50	1.63	1.91	1.28	17.00
2.38	7.00	26.38	4.44	1.00	2.13	1.50	1.91	1.28	23.68

NOTES: (APPLY TO SHEET 1 ONLY - SEE SHEET 10 FOR GENERAL NOTES):

1. THIS SHEET FOR TYPE I RADIATORS.
2. PROVIDE AIR SEAL STRIP ACROSS TOP AND BOTTOM OF CORE WHERE APPLICABLE TO PROVIDE FOR FLUSH SHROUD MOUNTING IN ACCORDANCE WITH "U" IF DISTANCE BETWEEN SEALING SURFACES EXCEEDS 0.13
3. PROVIDE HORIZONTAL AIR SEAL FILLER STRIP IF "U" EXCEEDS CORE HEIGHT TOLERANCE LIMITS.
4. PROVIDE 1/4 PLATE FLANGE COVER WITH 1/16 GASKET, 2 HEX. HD. CAP SCREWS, 3/8-16 UNC-2 X 3/4 LG, 2 LOCKWASHERS FOR EACH UNUSED FLANGE
5. FOR MS 35773-2 FLANGE, 5 REQUIRED.

APPROVED 9 JUNE 1958 REVISED (B) 8 APR 1966

(B) ENTIRE STANDARD REVISED AND REDRAWN

P.A. ARMY MO (ERDL) Other Cust YD 82	TITLE: RADIATORS, ENGINE COOLING, INDUSTRIAL	MILITARY STANDARD MS 35773
PROCUREMENT SPECIFICATION	SUPERSEDES	SHEET 1 OF 10

DD FORM 1 SEP 67 672-1 (Coordinated)

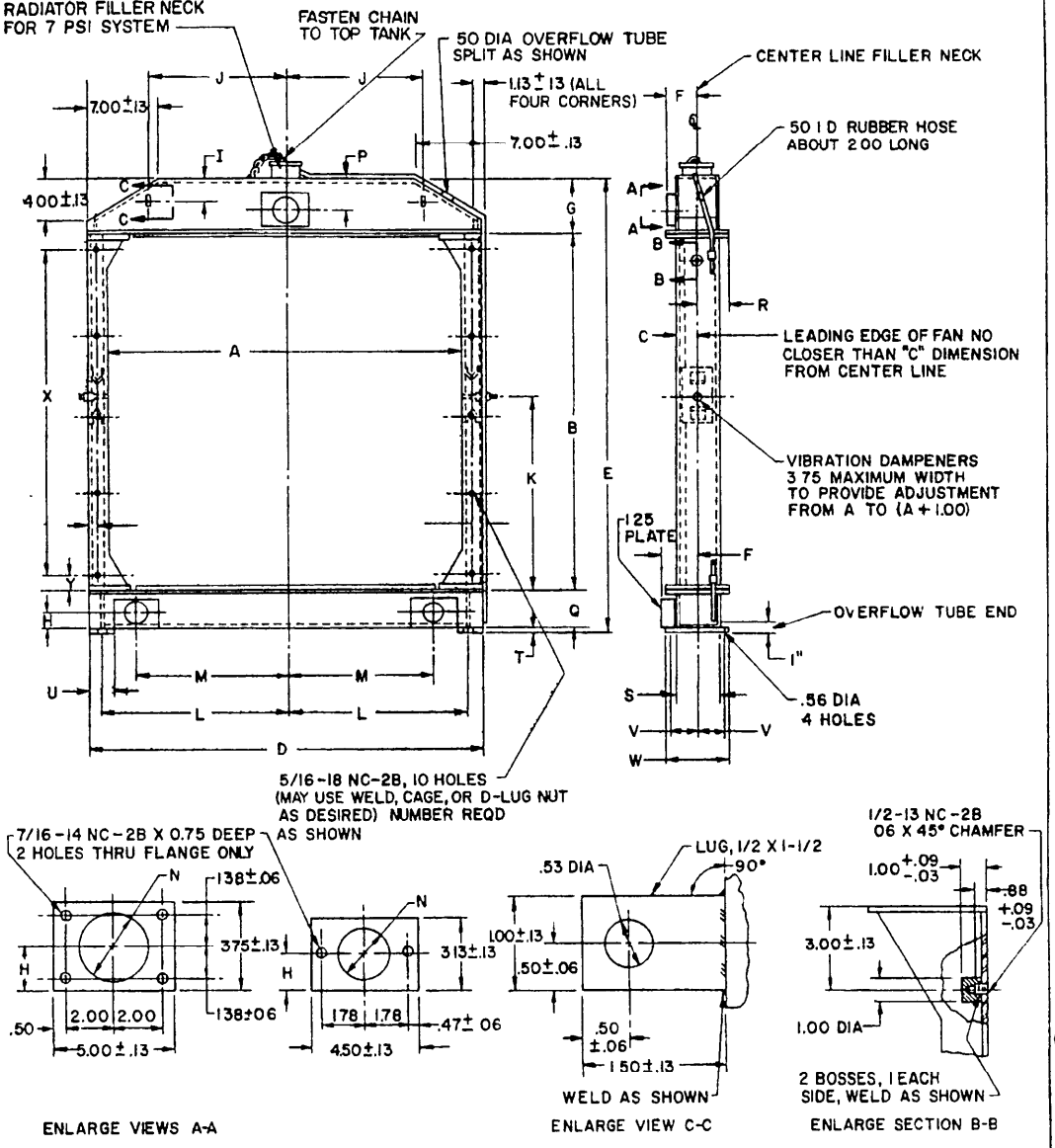
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MS-51062
RADIATOR PRESSURE CAP
MS-51063
RADIATOR FILLER NECK
FOR 7 PSI SYSTEM

FED SUP CLASS
2930



ENLARGE VIEWS A-A

ENLARGE VIEW C-C

ENLARGE SECTION B-B

NOTES (APPLY TO SHEET 2 ONLY)

6. THIS SHEET FOR TYPE II RADIATORS
7. FOR A THRU Y DIMENSIONS, SEE SHEET 3, FOR DETAIL OF CORE SEE SHEET 5
8. PROVIDE 1/4 PLATE FLANGE COVER WITH 1/16 GASKET AND HEX HD CAP SCREWS, 7/16-14 UNC-2 X 3/4 LONG, AND 2 OR 4 LOCKWASHERS, AS APPLICABLE, FOR ONE OF THE BOTTOM OUTLETS.
9. PROVIDE TOP TANK BAFFLE.

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P.A. ARMY MO (ERDL)	TITLE	MILITARY STANDARD
Other Cust YD 82	RADIATOR, ENGINE COOLING, INDUSTRIAL	MS 35773
PROCUREMENT SPECIFICATION MIL-R-45306	SUPERSEDES	SHEET 2 OF

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MS PART NO.	A ±.13	B ±.06	C ±.13	D ±.13	E ±.18	F ±.13	G ±.13	H ±.06	I ±.13	J ±.13	K ±.13 SEE NOTE 13	L ±.13
MS-35773-4	23.50	27.00	2.38	27.00	36.38 SEE NOTE 11	3.63	5.00	1.56	2.50	8.00	13.50	12.75
MS-35773-5	27.00	30.00	2.38	30.50	39.38 SEE NOTE 11	3.63	5.00	1.56	2.50	10.00	15.00	14.50
MS-35773-6	33.50	36.00	2.38	38.00	47.00 SEE NOTE 12	3.63	6.00	1.88	3.00	12.00	18.00	17.50
MS-35773-7	39.50	40.00	2.38	44.00	51.00 SEE NOTE 12	3.63	6.00	1.88	3.00	15.00	20.00	20.50
MS-35773-8	43.50	44.00	2.38	48.00	55.00 SEE NOTE 12	3.63	6.00	1.88	3.00	17.00	22.00	22.50

M ±.13	N ±.09	P ±.06	Q ±.13	R ±.13	S ±.13	T ±.06	U ±.13	V ±.13	W ±.13	X	Y ±.09	CORE MS PART NO.
9.50	2.18	2.50	4.00	3.50	4.75	0.38	1.50	3.00	7.00	4-HOLES @ 8"O.C = 24"	1.50	MS-35884-1
11.25	2.18	2.50	4.00	3.50	4.75	0.38	1.50	3.00	7.00	4-HOLES @ 9" O.C.=27"	1.50	MS-35884-2
13.50	3.00	3.38	4.50	3.50	4.75	0.50	3.00	3.00	7.00	5-HOLES @ 8"O.C = 32"	2.00	MS-35884-3
16.50	3.00	3.38	4.50	3.50	4.75	0.50	3.00	3.00	7.00	5-HOLES @ 9"O.C = 36"	2.00	MS-35884-4
18.50	3.00	3.38	4.50	3.50	4.75	0.50	3.00	3.00	7.00	5-HOLES @ 10"O.C =40"	2.00	MS-35884-5

NOTES: (APPLY TO SHEET 3 ONLY):
10. THIS SHEET FOR TYPE II RADIATORS.
11 INCLUDES 38" THICK MOUNTING FOOT
12 INCLUDES .50" THICK MOUNTING FOOT.
13 BOTTOM OF SIDE SUPPORT TO CENTER LINE OF VIBRATION DAMPENER

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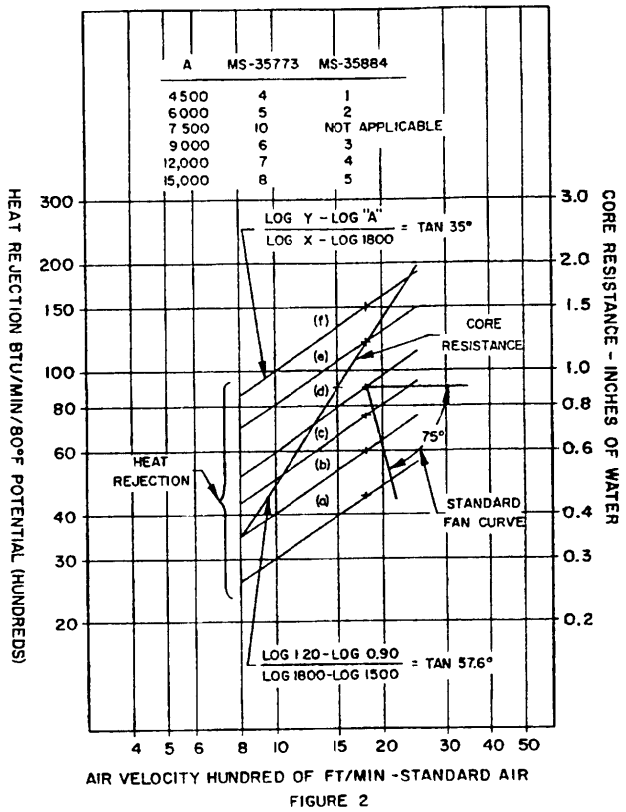


FIGURE 2

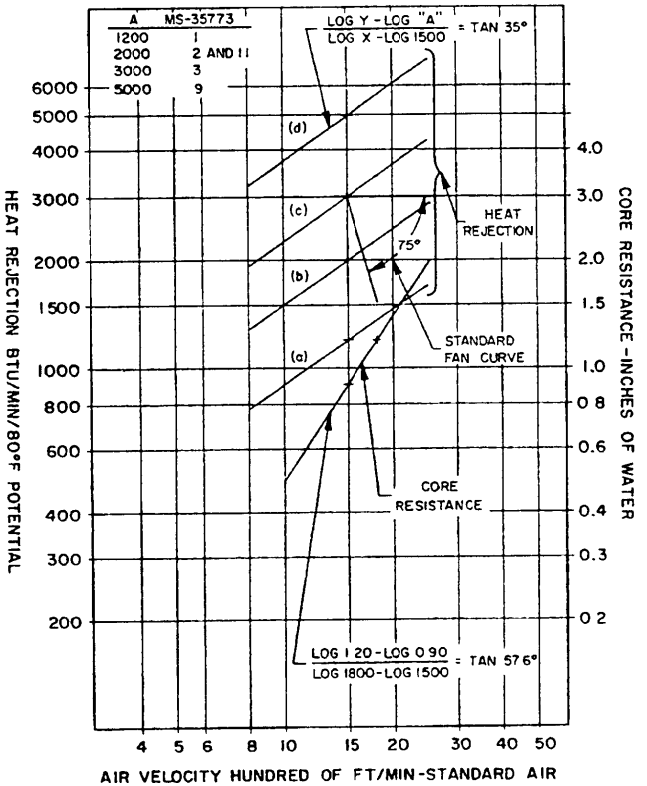


FIGURE 1

NOTES (APPLY TO SHEET 4 ONLY):

14 FOR FIELD TESTING, THE TOP TANK TEMPERATURE MINUS ENTERING AIR TEMPERATURE SHOULD NOT EXCEED 85°F. THUS, IN A 125°F AMBIENT TEMPERATURE THE TOP TANK WOULD RUN 210°F MAX

15. CORES SHALL MEET THE TEST REQUIREMENTS WHEN COMPARED ON THE BASIS OF THE STANDARD FAN CURVE OF FIGURES 1 & 2 WITHIN THE FOLLOWING LIMITATIONS
   A HEAT REJECTION - EQUAL TO OR GREATER THAN RATED
   B AIR VELOCITY - RATED PLUS 12 PERCENT

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FED SUP CLASS 2930

MILITARY STANDARD

MS 35773

SHEET 4 OF

TITLE: RADIATORS, ENGINE COOLING, INDUSTRIAL

SUPERSEDES

PROCUREMENT SPECIFICATION MIL-R-45306

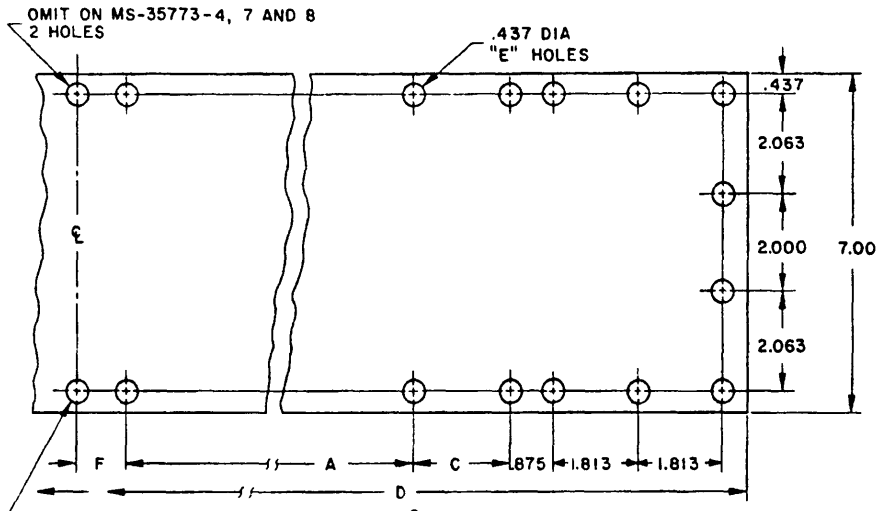
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Other Cust YD 82

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BOLT HOLES ARE SYMMETRICAL ABOUT ℄ OF HEADER PLATE AS SHOWN
PROVIDE FOR SCREW, CAP HEX HD, 3/8-16 NC X 1-1/4 LG, CAD PL STL,
AND NUT, PLAIN, HEX., 3/8, WITH LOCKWASHER

RADIATOR MS PART NO	A	C	D ±.06	E	F	CORE MS PART NO.
MS-35773-4	4 HOLES @ 2" OC = 6"	1.500	27.00	36	1.000	MS-35884-1
MS-35773-5	4 HOLES @ 2" OC = 6"	2.250	30.50	38	2.000	MS-35884-2
MS-35773-6	6 HOLES @ 2" OC = 10"	2.000	38.00	46	2.000	MS-35884-3
MS-35773-7	8 HOLES @ 2" OC = 14"	2.000	44.00	52	1.000	MS-35884-4
MS-35773-8	9 HOLES @ 2" OC = 16"	2.000	48.00	56	1.000	MS-35884-5

NOTES (APPLY TO SHEET 5 ONLY)

16. THIS SHEET FOR TYPE II RADIATORS

17 UNLESS OTHERWISE SPECIFIED ALL DECIMALS ARE ±.03 FOR TWO PLACE AND ±.010 FOR THREE PLACE

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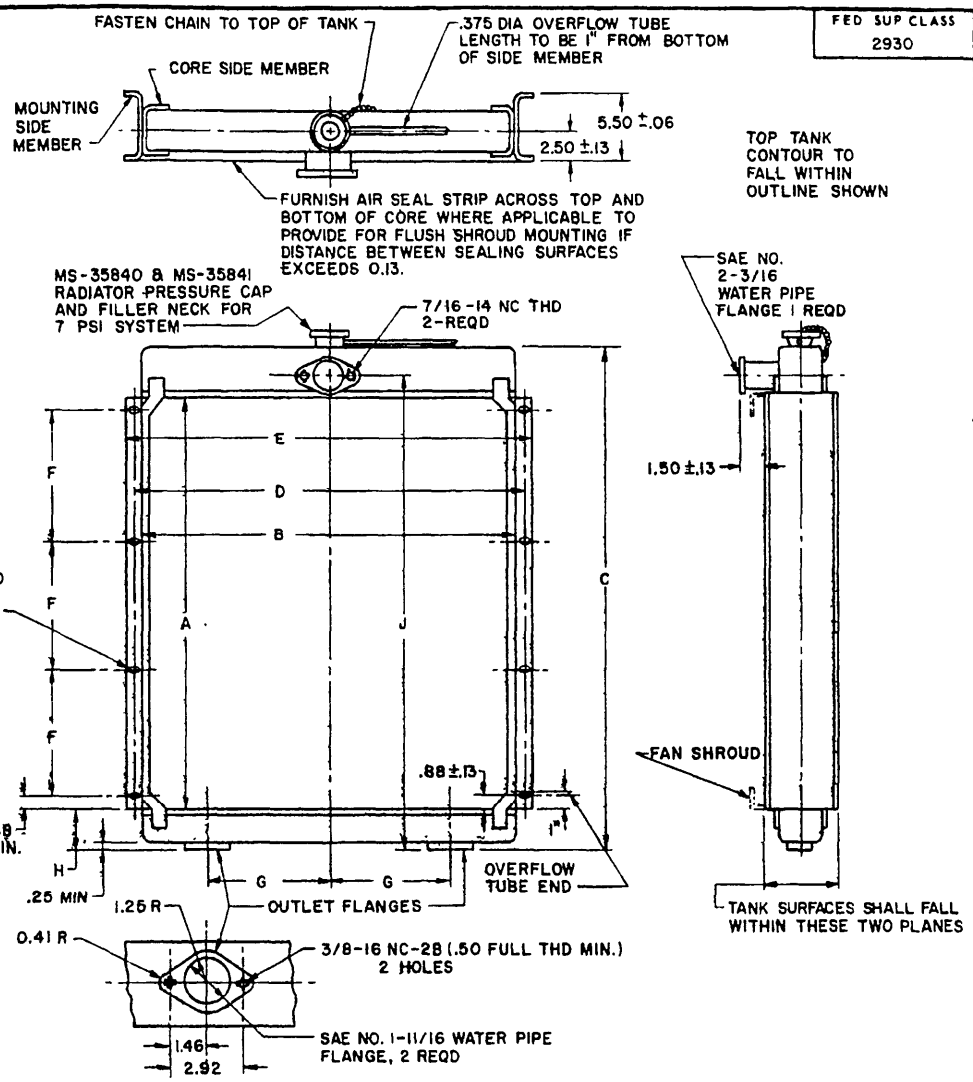
P.A. ARMY MO (ERDL) Other Cust YD 82	TITLE RADIATORS, ENGINE COOLING, INDUSTRIAL	MILITARY STANDARD MS 35773
PROCUREMENT SPECIFICATION MIL-R-45306	SUPERSEDES	SHEET 5 OF

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REFERENCE DIMENSIONS ONLY

MS PART NO.	A ±.13	B MAX	C ±.13	D ±.13	E ±.13	F	G ±.13	H ±.09	J ±.13
MS-35773-9	28.00	29.00	35.06	30.75	32.00	8.75	9.00	3.94	32.81
MS-35773-10	34.00	31.50	41.56	32.75	34.00	10.75	10.50	4.44	39.31

NOTES: (APPLY TO SHEET 6 ONLY)

18. THIS SHEET FOR TYPE I RADIATORS.
19. PROVIDE 1/4 PLATE FLANGE COVER WITH 1/16 GASKET, 2 HEX. HD CAP SCREWS, 3/8-16 UNC-2 X 3/4 LONG, 2 LOCKWASHERS FOR ONE OF THE BOTTOM OUTLETS
20. PROVIDE TOP TANK BAFFLE.

APPROVED 9 JUNE 1958 REVISED (B) For Changes See Sheets 1 thru 10

P.A. ARMY MO (ERDL) Other Cust YD 82	TITLE RADIATORS, ENGINE COOLING, INDUSTRIAL	MILITARY STANDARD MS 35773
PROCUREMENT SPECIFICATION MIL-R-45306	SUPERSEDES.	SHEET 6 OF

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HEAT REJECTION AND CORE RESISTANCE REQUIREMENTS FOR MILITARY STANDARD RADIATORS

LEGEND SEE NOTE 21	MS PART NUMBER	HEAT REJECTION BTU/MIN ("A") *MIN REQ ALLOWABLE	AIR VELOCITY FT/MIN **RATED VALUE	MAX RESISTANCE INCHES WATER ***MAX REQ ALLOWABLE	WATER FLOW GALS/MIN (RATED VALUE)
1(a)	MS-35773-1	* 1200 1363 1519	**1500 1800 2100	***0.9 1.2 1.5	15
1(b)	MS-35773-2 MS-35773-11	* 2000 2272 2531	**1500 1800 2100	***0.9 1.2 1.5	25
1(c)	MS-35773-3	* 3000 3408 3797	**1500 1800 2100	***0.9 1.2 1.5	38
1(d)	MS-35773-9	* 5000 5682 6328	**1500 1800 2100	***0.9 1.2 1.5	62
2(a)	MS-35773-4 MS-35884-1	3960 * 4500 5012	1500 **1800 2100	0.9 ***1.20 1.5	56
2(b)	MS-35773-5 MS-35884-2	5281 * 6000 6684	1500 **1800 2100	0.9 ***1.20 1.5	75
2(c)	MS-35773-10	6600 * 7500 8355	1500 **1600 2100	0.9 ***1.20 1.5	92
2(d)	MS-35773-6 MS-35884-3	7921 * 9000 10020	1500 **1800 2100	0.9 ***1.20 1.5	110
2(e)	MS-35773-7 MS-35884-4	10560 * 12000 13370	1500 **1800 2100	0.9 ***1.20 1.5	150
2(f)	MS-35773-8 MS-35884-5	13200 * 15000 16710	1500 **1800 2100	0.9 ***1.20 1.5	185

NOTES (APPLY TO SHEET 7 ONLY):

21. REFER TO FIGURES AND CURVES - SHEET 4.

22. ABOVE REQUIREMENTS ARE BASED ON 80° F POTENTIAL.
80° F POTENTIAL = AVERAGE WATER - ENTERING AIR

AVERAGE WATER = $\frac{T_1 + T_2}{2}$; T_1 = TOP TANK TEMPERATURE, T_2 = BOTTOM TANK TEMPERATURE

HEAT REJECTION CURVE = $\frac{\text{LOG } Y - \text{LOG "A"}}{\text{LOG } X - \text{LOG } 1500}$ = TAN 35°

RESISTANCE CURVE = $\frac{\text{LOG } 1.20 - \text{LOG } 0.90}{\text{LOG } 1800 - \text{LOG } 1500}$ = TAN 57.6°

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PROCUREMENT SPECIFICATION MIL-R-45306	SUPERSEDES	SHEET 7

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MATERIAL REQUIREMENTS

RADIATOR TYPE	MS PART NO	TANKS (SEE NOTE 23)	SIDES	HEADER PLATES		HEADER BARS (SEE NOTE 23)	NOMINAL TUBE WALL THK	NOMINAL END LIVE TUBE WALL THK
				TOP	BOTTOM			
I	MS 35773-1	#20 GA. (.032) BRASS	#18 GA. (.047) TERNE WITH #14 GA. (.080) SAE 1010 WING	#20 GA. (.032) BRASS	#20 GA (.032) BRASS	-----	.006	.008
I	MS 35773-2 MS 35773-11					-----	.006	.008
I	MS 35773-3					-----	.006	.008
I	MS 35773-9					-----	.006	.008
I	MS 35773-10	#18 GA. (.040) BRASS	#10 GA. (.135) SAE 1010	#18 GA (.040) BRASS	#18 GA (.040) BRASS	-----	.006	.008
II	MS 35773-4 MS 35884-1	#10 GA (.135) SAE 1010				SAE 1010	.006	.008
II	MS 35773-5 MS 35884-2				#11 GA (.091) BRASS	SAE 1010	.008	.010
II	MS 35773-6 MS 35884-3	#7 GA (.177) SAE 1010	#7 GA (.177) SAE 1010			SAE 1010	.008	.010
II	MS 35773-7 MS 35884-4					SAE 1010	.008	.010
II	MS 35773-8 MS 35884-5					SAE 1010	.008	.010

NOTES (APPLY TO SHEET 8 ONLY).

23 NOT APPLICABLE TO STANDARD MS 35884.

24 THE USE OF COPPER BEARING STEEL OR INGOT IRON IS ALLOWED IN PLACE OF SAE 1010

25. INTERNAL SURFACES OF PARTS CONSTRUCTED FROM SAE 1010 SHALL BE PROTECTIVE COATED

26. GAGE SIZES SHOWN ARE MINIMUM.

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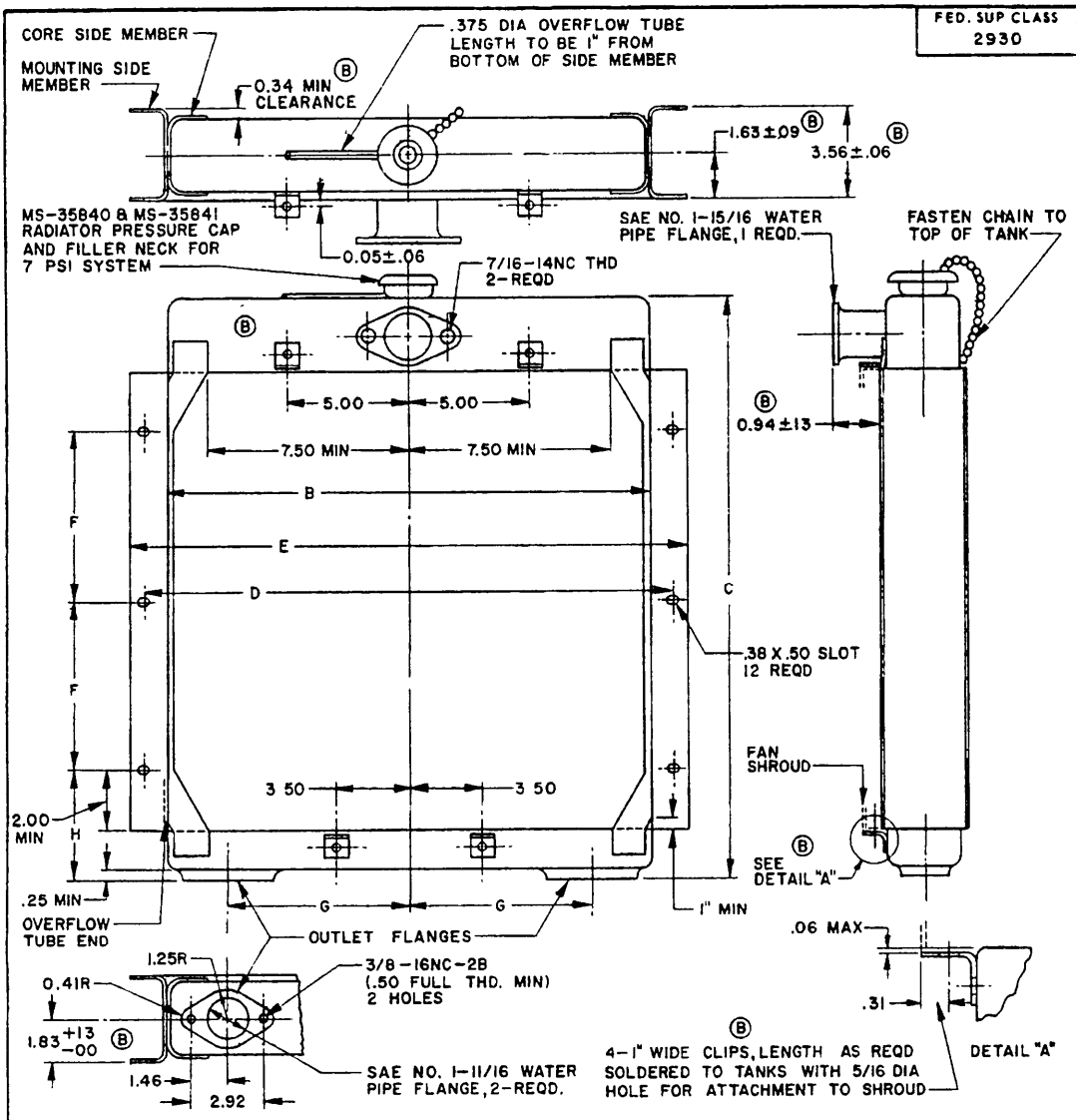
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REFERENCE DIMENSIONS ONLY

MS PART NO.	A ±.13	B MAX	C ±.13	D ±.13	E ±.13	F	G ±.13	H ±.09	J ±.13
MS-35773-11	19.00	20.25	23.88	22.00	23.00	7.00	7.50	4.38	22.50

NOTES (APPLY TO SHEET 9 ONLY)

26. THIS SHEET FOR TYPE I RADIATORS.
27. PROVIDE 1/4 PLATE FLANGE COVER WITH 1/16 GASKET, 2 HEX. HD CAP SCREWS, 3/8-16 UNC-2 X 3/4 LONG, 2 LOCKWASHERS FOR ONE OF THE BOTTOM OUTLETS
28. PROVIDE FOR TOP TANK BAFFLE
29. PROVIDE HORIZONTAL AIR SEAL FILLER STRIP IF "A" EXCEEDS CORE HEIGHT TOLERANCE LIMITS.

APPROVED 9 JUNE 1958 REVISED (B) For Changes See Sheets 1 thru 10

P.A ARMY MO(ERDL) Other Cust YD 82	TITLE: RADIATORS, ENGINE COOLING, INDUSTRIAL	MILITARY STANDARD MS 35773
PROCUREMENT SPECIFICATION MIL-R-45306	SUPERSEDES	SHEET 9 OF

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GENERAL NOTES (APPLY TO SHEETS 1 THRU 9):

30. UNLESS OTHERWISE SPECIFIED, TOLERANCES ON DECIMAL DIMENSIONS ± .03 INCHES.

31. RADIATORS MS 35773-1 THRU MS 35773-11 WILL BE TESTED IN WATER WITH AIR PRESSURE AT 8 LBS. MINIMUM TO 10 LBS MAXIMUM PRESSURE

32. ALL DIMENSIONS ARE IN INCHES.

33. FOR DESIGN FEATURE PURPOSES, THIS DOCUMENT TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN

34. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON THE DATE OF INVITATION FOR BIDS.

(B) SHEET ADDED

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P.A. ARMY MO (ERDL) Other Cust YD 82	TITLE RADIATORS, ENGINE COOLING, INDUSTRIAL	MILITARY STANDARD MS 35773
PROCUREMENT SPECIFICATION MIL-R-45306	SUPERSEDES.	SHEET 10 OF

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